TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: **Duration time**: from 09/2026 To 03/2027

(*) Profession: Software Engineer **Specialty**: SE [x]

(*) Kinds of person make registers: Lecturer [x] Students

1. Register information for supervisor (if have)

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3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

3.1.1. English: ***StockFlow Commerce – Unified Warehouse & E-commerce Platform for F&B Supplies with Custom Cup & Packaging Printing***

3.1.2. Vietnamese: ***Nền tảng Kho vận & Thương mại điện tử cho ngành F&B tích hợp In ly và bao bì theo yêu cầu***

(*) 3.2. Main proposal content (including result and product)

a) Context:

Vietnam's beverage retail sector (milk tea shops, coffee chains and takeaway kiosks) uses large quantities of disposable cups and packaging such as plastic cups, paper cups and takeaway bags. Many of these items are printed with the shop's own logo for branding. These products are supplied by specialized B2B distributors. Their customers are beverage shops and

chains that buy in bulk, reorder regularly, and often place custom logo-printing orders that the supplier produces in its own print workshop.

In practice, these suppliers usually run two separate worlds. The warehouse side handles purchasing from manufacturers, receiving and quality checking, storage, picking, packing, shipping and stock transfers between warehouses. It is often managed with spreadsheets, paper slips or a standalone warehouse tool. Taking orders from business customers is handled through phone calls, chat apps and manual quotations. Because the two sides share no data, several problems keep coming up: Inventory is not reliable. Stock levels quoted to business customers lag behind what is actually in the warehouse. This leads to overcommitted orders, late deliveries and manual cross-checking. Product data is duplicated. The same cup or packaging item is recorded separately for warehouse and sales use, often in different units (piece, sleeve, carton). This causes errors in bulk-order quantities and pricing. There is no end-to-end traceability. Nothing links a supplier purchase order to the received batch, its storage location, the customer order and the outgoing shipment. Disputes, returns and supplier invoice checks therefore take a long time. Warehouse operations are inefficient. Without a warehouse map, putaway suggestions or optimized picking routes, staff rely on memory to find goods. This increases travel distance and picking mistakes, and the problem grows with larger order volumes and more warehouses. Custom printing is slow and error-prone. Shop owners send logos through chat. Designs go back and forth many times before approval, and the final artwork is passed to the print workshop by hand. Customers cannot see how the printed cup will look before they confirm the order. Account communication is scattered. Advice, design approval and order changes for each business customer happen in personal chat apps that are not linked to orders. Information is lost when staff change, and managers cannot track it.

These problems call for one integrated platform. It should combine a Warehouse Management System (WMS) with a B2B e-commerce platform that share a single product master and real-time inventory. On the platform, business customers can design and preview their own printed packaging, and that design flows directly into in-house production and fulfillment. StockFlowCommerce is proposed for this context. It connects the full chain: procurement → receiving → warehousing → online B2B ordering and payment → print production → packing → shipping. It also supports sales staff and business customers through a built-in chat channel.

b) Proposed Solutions:

StockFlowCommerce is an integrated platform that combines a Warehouse Management System (WMS) for back-office operations with a B2B e-commerce platform for business customers. It is built for suppliers of disposable cups and beverage packaging that sell plain items from stock and also print customers' logos to order in their own workshop.

Both parts of the system share one product master, so each item is defined once. Unit-of-measure conversion (piece, sleeve, carton) is built in, and all stock is recorded in a single base unit. Inventory is tracked by bin and storage area across multiple warehouses. Every stock movement must come from a source document (purchase order, receipt, pick, transfer, adjustment). This gives end-to-end traceability, from the supplier purchase order to the

shipment delivered to the customer. At checkout, the system reserves stock for the order so that a business customer cannot order more than is actually available.

To make warehouse work faster and more accurate, the system provides: an interactive 2D/3D warehouse map, a slotting engine that suggests where to store goods, optimized picking routes, 3D load planning for full-truckload shipments

On the sales side, business customers can design their logo on cups and packaging in a 2D editor with a live 3D preview, and approve the design before payment. The approved design is attached to the order unchanged and goes straight into an in-house production order. Production covers printing, QC, packing and shipping. A built-in sales chat links each conversation with sales staff to the customer's orders, so advice, design approval and changes to an order happen in one place that can be tracked.

c) Functional Requirements:

Product Management

- Create product & variants: Create a product under a category and define its variant attributes (capacity, material, size). The system generates a unique SKU for each variant.
- Manage inventory item information: Maintain unit of measure (UoM) and UoM conversions (e.g., 1 carton = 1,000 pcs), weight, dimensions, storage class, lot/expiry tracking and stock thresholds for each SKU.
- Manage sales information: Maintain the description, images, sales units and a selling price for each sales unit.
- Approve & publish product: Submit, approve or reject products, publish them to the storefront, or discontinue them.

Procurement

- Generate replenishment proposal: Automatically list the SKUs that have fallen below their reorder point.
- Create purchase order (PO): Create a PO with supplier, receiving warehouse, expected date and line items, with automatic unit conversion and total calculation.
- Approve & confirm PO: Send POs above the limit for approval. A confirmed PO is locked and can only be changed through an amendment.
- Track & close PO: Track received and outstanding quantities, then close or cancel the PO.

Goods Receipt & Quality Control

- Receive goods: count goods against the PO, recording lot, expiry and serial numbers.
- Handle receiving discrepancies: Record short, over or wrong-SKU deliveries
- Quality inspection: Classify received goods as Accepted, Quarantine or Rejected. Only accepted goods move on to storage.
- Receive returned goods: Receive customer returns and failed deliveries, inspect them, and restock or scrap them.

Vendor Invoice Matching

- Record vendor invoice: Enter or import a supplier invoice and link it to the PO and goods receipt.
- Three-way matching: Compare PO, receipt and invoice for quantity, price and tax, and flag any variances.

Putaway

- Generate putaway tasks: Automatically create tasks to store goods that have been received and passed QC.
- Storage Location Recommendation: Recommend and rank suitable storage bins based on storage constraints, available capacity, proximity to the packing area, FIFO/FEFO policies, and space utilization.
- Confirm putaway: Scan the bin to confirm storage, with constraint validation and a mandatory reason when the suggestion is overridden.

Warehouse Map 2D/3D

- Design warehouse layout: Define walls, doors, functional areas, shelves, shelf levels and bins
- Generate location codes: Generate a unique location code and barcode for every bin and area.
- View interactive map: View the warehouse in 2D/3D with a fill-rate heatmap and location search.
- Optimize storage locations (re-slotting): Analyze storage locations and propose moves to put fast-moving items closer to the packing area.

Inventory Control

- Track inventory by location: Track stock by SKU, lot and location across multiple warehouses.
- Manage stock status: Track stock statuses (Inbound, Available, Allocated, Quarantine, In-Transit) and calculate the quantity available to sell (ATP).
- Cycle count & adjustment: Count stock at locations and adjust stock with manager approval.
- Low-stock alerts: Send alerts when stock falls below the minimum threshold.

Stock Transfer

- Transfer within a warehouse: Move stock between locations in the same warehouse with scan confirmation.
- Transfer between warehouses: Create, approve, ship and receive transfer orders between warehouses, tracking stock while it is in transit.

Picking

- Allocate stock: Assign each order to specific bins and lots by FEFO
- Plan picking waves & routes: Group orders into picking waves and optimize the picking route on the warehouse map.
- Execute picking: Scan location, SKU and quantity to confirm each pick.
- Handle short picks: Re-allocate from other locations, or switch to backorder or partial delivery when stock runs short.

Production (Print-on-Demand)

- Generate production orders: Assignee task a production order for each custom-print order line, with the confirmed design attached.

- Check print file: Review the print file before production and put the order on hold if the file has errors.
- Schedule production: Queue production orders by due date, assign them to print stations, and estimate completion dates.
- Report production & QC: Record good and defective quantities, check finished goods against the design, and reprint any shortfall.

Packing

- Verify order: Scan to check that the items and quantities match the order before packing.
- Pack & create shipping label: Record package weight and dimensions, get a tracking number and print the shipping label.

Shipping

- Hand over to carrier: Create shipments, scan packages at handover and print the manifest.
- Plan 3D truck loading: Calculate the number of trucks needed and where each package goes, and show the loading plan in 3D.

Browse & Catalog

- Manage catalog: Manage categories, display order, featured products, prices and promotions.
- Browse & search products: Browse by category and search or filter by product group, capacity, material and price.
- View product details: View images, description, variants, price per sales unit and real-time availability.
- Wishlist: Save products to a wishlist

Product Design 2D/3D

- Design in 2D: Upload a logo, add text , and arrange the layout within the print area.
- Live 3D preview: See the design rendered in real time on a 3D model of the product.
- Save & confirm design: Save design drafts and versions, then confirm the final design to add it to the cart.

Cart & Checkout

- Manage cart: Add, update or remove items, change the sales unit and apply vouchers.
- Checkout: Select the shipping address, tax and order total.
- Reserve stock: Reserve stock when the order is placed and release it if the order is cancelled or payment fails.

Payment

- Pay online: Pay through a payment gateway,
- Deposit: Pay a deposit for large or custom-print orders and pay the balance later.
- Refund: Refund an order fully or partially.

Order Management

- Track orders: Search orders and view their status timeline from placement to delivery.
- Select fulfillment warehouse: Choose the warehouse that fulfills the order based on stock, distance and printing capability, and split orders across warehouses when needed.
- Amend orders: Change address, quantities or items, depending on the current order status.

- Cancel orders: Cancel an order, stop the related warehouse tasks and refund the payment.
- Returns & exchanges: Handle return and exchange requests, receive the returned goods and issue refunds.

Sales Chat

- Chat with sales: Chat in real time with sales staff, with the current product, design or order attached as context.

Customer Account

- Register & log in: Register with email or phone number and verify it.
- Manage profile & addresses: Update personal information and the shipping address book.
- Account security: Change password, manage login sessions.
- Order & design history: View past orders, saved designs and chat history.

System Administration

- Manage users & roles: Create staff accounts and assign roles and approval limits.
- Manage warehouses: Register multiple warehouses, each with a unique code prefix.
- Approval workflow: Approve or reject documents.
- Notifications: Send email, SMS and in-app notifications for key events.
- Audit log: Record every change to stock and documents.

d) Non-Functional Requirements:

Performance Efficiency

- The system shall respond to common read/write operations within 2 seconds under the expected demo workload (≤ N concurrent users).
- The system shall process cross-application tasks asynchronously so that user-facing requests are not blocked by background work.
- Each of the three applications shall be independently deployable and horizontally scalable without requiring redeployment of the others.

Reliability

- The system shall maintain a single authoritative source of inventory data; any read replica used by the storefront shall converge with the source within 2 seconds of a stock change.
- The system shall guarantee that no order is confirmed for stock that is unavailable (zero oversell), even under concurrent checkout.
- The system shall preserve a complete, immutable history of stock and payment movements, with no transaction data lost during state transitions.

Security

- The system shall enforce role-based access control such that each actor can only perform operations authorized for their role.
- The system shall restrict access to internal warehouse operations to whitelisted network addresses.
- The system shall authenticate requests using stateless, asymmetrically-signed access tokens; refresh tokens shall be stored only in hashed form, rotated on each use, and revocable.

- The system shall encrypt personal and payment data in transit and at rest, and shall process payment-gateway callbacks idempotently.

Usability

- The system shall provide a Vietnamese-language interface across both the public storefront and the internal console.
- The internal console shall present menus and actions filtered according to the user's role.
- The interface shall render correctly on both desktop and mobile web viewports.

Maintainability & Portability

- The system shall be structured into independent modules with explicitly defined boundaries, coupled only through a shared product identifier (SKU) and asynchronous events.
- The system shall expose documented APIs, enabling individual modules to be extracted into standalone services with minimal changes to other modules.

e) Theory & Practical:

- User Requirement Specification (URS)
- Software Requirement Specification (SRS)
- Architecture Design Document (ADD)
- Detailed Design Document (DDD)
- System Implementation Document
- Testing Document (Unit, Integration, System, UAT)
- Deployment & Installation Guide
- Source code and deployable software packages

f) Products (Expected Deliverables):

Web API for System: A complete set of RESTful APIs covering all business processes of the unified WMS and e-commerce platform:

- Storefront: browse catalog, design products, cart and checkout, payment.
- Procurement: replenishment proposal, purchase order, receipt and QC, three-way invoice matching, putaway.
- Warehouse operations: warehouse map and slotting, picking, packing, shipping and 3D load planning, transfers within a warehouse and between warehouses.
- In-house print-on-demand production.
- Order lifecycle management, customer data and sales chat.

g) Client Interfaces

Customer (Storefront): Browse and search the cup and packaging catalog. Design logo prints in a 2D editor and see them on a live 3D model. Manage the cart and place orders . Pay online with a deposit. Track orders and shipments, chat with Sales, request changes, cancellations or returns, and manage the profile, address book and saved designs.

Sales & Order Operations (Sales Staff, Order Coordinator, E-commerce Admin): Advise customers through context-aware chat (linked to a product, design, cart or order). Finalize

designs, create orders for customers and send payment links. Release confirmed orders to a warehouse, handle pending orders, and split or merge orders. Manage catalog, pricing, promotions and design templates, and run prepress checks on print files before production.

Warehouse & Production Operations (Procurement Staff, Inventory Planner, Warehouse Staff, Warehouse Manager, QC Staff, Production Staff, Accountant): Turn replenishment proposals into purchase orders, receive goods in 2-step or 3-step flows (with QC), and put them away using slotting suggestions on the 2D/3D warehouse map. Pick, pack and hand over to carriers, and plan 3D truck loading for full-truckload shipments. Run print orders on print stations and record QC of finished goods. Run transfers within and between warehouses. Reconcile purchase order, receipt and invoice,

System Admin: Configure warehouses (warehouse prefix, layout of shelves, levels, bins and areas), manage users and RBAC, maintain product master data (Product/Variant ↔ Inventory Item, units of measure, customization settings), and configure slotting rules and system parameters.

h) Proposed Tasks:

Task Package 1 – Core Platform, Storefront Commerce & Order Orchestration:

- User management, RBAC, multi-warehouse configuration, and product master data with unit-of-measure conversion.
- Catalog with search and filters scoped to manufacturable products only.
- Cart and checkout with stock reservation.
- Payment coordination: online gateway, deposit, refund.
- Order lifecycle state management
- Customer profile management and real-time sales chat.

Task Package 2 – Warehouse Operations & Real-time Inventory:

- Procurement flow: replenishment proposal → purchase order → multi-step receipt with QC/quarantine → three-way invoice matching → putaway.
- 2D/3D warehouse map with a rule-based slotting engine.
- Picking with route optimization, packing, and shipping through carrier API integration.
- Full-truckload shipping with 3D load planning.
- Transfers between warehouses (in-transit tracking) and within a warehouse (replenishment, re-slotting).
- Location-based inventory where every change is backed by a source document.

Task Package 3 – Product Customization & In-house Production:

- 2D product designer with live 3D preview and an immutable design snapshot attached to each order line.
- Production orders, print station queues and capacity-based completion estimates.
- Issue blank stock (unprinted cups/packaging) from bins to the print shop, QC finished goods (with a rework limit), and hand them straight to packing.

Task Package 4 – Testing, Deployment & Documentation:

- Test the entire system (Unit, Integration, System, UAT), focusing on inventory consistency under concurrent orders and the state transitions of each document type.
- Deploy a demo environment with sample data (warehouses, SKUs, orders).
- Prepare complete documentation for the defense panel: BRD, SRS, architecture and database design, API specification, test reports and user guides.

4. Other comments (propose all relative things if have):

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HCM, date 07/08/2026

Supervisor (If have)	**On behalf of Registers**
(Sign and full name)	*(Sign and full name)*